

Arithmetic Functions I:

Elementary Theory

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Chapter 0 Primes and the Fundamental Theorem of Arithmetic

Chapter 1 Arithmetic functions I: Elementary theory

The Moebius function

The Euler phi (totient) function

The von Mangoldt function

The divisor and sum-of-divisors functions

Dirichlet product

Definition 1 (Divisibility).

Given two integers $d \neq 0$ and n , we say that d **divides** n or n is **divisible by** d , if there exists an integer m such that $n = dm$. We write $d|n$ if d divides n , and $d \nmid n$ if d does not divide n .

Theorem 1 (Division Algorithm).

Given any $a \in \mathbb{N}$ and $b \in \mathbb{Z}$, there exist unique integers q and r that satisfy

$$b = aq + r \text{ and } 0 \leq r < a. \tag{1}$$

Definition 2.

The **greatest common divisor (gcd)** of two integers a and b that are not both zero is the unique integer $d > 0$ satisfying (i) $d|a$ and $d|b$, and (ii) if $c|a$ and $c|b$, then $c|d$. The gcd of a and b is denoted by (a, b) . If $(a, b) = 1$, then a and b are said to be **relatively prime** or **coprime**.

The **least common multiple (lcm)** of two non-zero integers a and b is the unique integer $m > 0$ satisfying (i) $a|m$ and $b|m$, and (ii) if $a|n$ and $b|n$, then $m|n$. The lcm of a and b is denoted by $[a, b]$.

The Fundamental Theorem of Arithmetic

Theorem 2 (Fundamental Theorem of Arithmetic).

Every integer $n > 1$ can be written as a product of primes, and the representation is unique up to the order of the factors.

Notations $n = p_1 \dots p_r$, p_i primes.

$n = p_1^{\alpha_1} \dots p_r^{\alpha_r}$, p_i distinct primes.

$n = \prod_p p^{\alpha(p)}$.

$n = \prod_{p^m \parallel n} p^m$, p^m exact power of p that divides n .

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Definition 3.

A function of the form $f : \mathbb{N} \rightarrow \mathbb{C}$ is called an *arithmetic function*.

1.

Moebius function: $\mu(n)$, defined by $\mu(1) = 1$, $\mu(n) = 0$ if n is not squarefree (i.e., divisible by the square of a prime), and $\mu(n) = (-1)^k$ if n is composed of k *distinct* prime factors (i.e., $n = \prod_i^k p_i$).

2.

Euler phi (totient) function: $\phi(n)$, the number of positive integers $m \leq n$ that are relatively prime to n ; i.e., $\phi(n) = \sum_{m=1, (m,n)=1}^n 1$.

3.

Von Mangoldt function: $\Lambda(n)$, defined by $\Lambda(n) = 0$ if n is not a prime power, and $\Lambda(p^m) = \log p$ for any prime power p^m .

Definition. An arithmetic function f is called **multiplicative** if $f \not\equiv 0$ and

$$(1.1) \quad f(n_1 n_2) = f(n_1) f(n_2) \quad \text{whenever } (n_1, n_2) = 1;$$

f is called **additive** if it satisfies

$$(1.2) \quad f(n_1 n_2) = f(n_1) + f(n_2) \quad \text{whenever } (n_1, n_2) = 1.$$

If this condition holds without the restriction $(n_1, n_2) = 1$, then f is called **completely (or totally) multiplicative** resp. **completely (or totally) additive**.

example, for fixed $k \in \mathbb{R}$, $r_k(n) := n^k$ is (completely) multiplicative.

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Characterization f is multiplicative iff $f(1) = 1$ and $f(n) = \prod_{p^m \parallel n} f(p^m)$.

Properties Assume f and g are multiplicative functions. The product $fg(n) := f(n)g(n)$ and f/g is also multiplicative.

Function	value at n	value at p^m	properties
$e(n)$	1 if $n = 1$, 0 else	0	unit element w.r.t. Dirichlet product, $e * f = f * e = f$
$\text{id}(n)$ (identity function)	n	p^m	
$s(n)$ (char. fct. of squares)	1 if $n = m^2$ with $m \in \mathbb{N}$, 0 else	1 if m is even, 0 if m is odd	
$\mu^2(n)$ (char. fct. of squarefree integers)	1 if n is squarefree, 0 else	1 if $m = 1$, 0 if $m > 1$	
$\mu(n)$ (Moebius function)	1 if $n = 1$, $(-1)^k$ if $n = \prod_{i=1}^k p_i$ (p_i distinct), 0 otherwise	-1 if $m = 1$, 0 if $m > 1$	$\sum_{d n} \mu(d) = 0$ if $n \geq 2$ $\mu * 1 = e$
$\lambda(n)$ (Liouville function)	1 if $n = 1$, $(-1)^{\sum_{i=1}^k \alpha_i}$ if $n = \prod_{i=1}^k p_i^{\alpha_i}$	$(-1)^m$	$\sum_{d n} \lambda(d) = n^2$ $\lambda * 1 = s$

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Theorem 3 (Moebius identity).

For all $n \in \mathbf{N}$, $\sum_{d|n} \mu(d) = e(n)$; i.e., the sum $\sum_{d|n}$ is zero unless $n = 1$, in which case is 1.

$\mu(1) = 1$, $\mu(2) = \mu(1 \times 2) = (-1)^2 = 1$, $\mu(3) = 1$, $\mu(4) = \mu(2 \times 2) = 0$, $\mu(30) = \mu(2 \times 3 \times 5) = -1$.

Application: Evaluation of sums involving a coprimality condition.

$$\begin{aligned} \sum_{n,(n,k)=1} f(n) &= \sum_n f(n) e((n,k)) = \sum_n f(n) \sum_{d|(n,k)} \mu(d) \\ &= \sum_{d|k} \mu(d) \sum_{n,d|n} f(n) = \sum_{d|k} \mu(d) \sum_m f(dm.) \end{aligned} \tag{2}$$

Evaluation of the Euler phi function $\phi(n) := \sum_{m \leq n, (m,n)=1} 1$. Eliminating the coprimality condition as above, yields the identity.

$$\phi(n) = \sum_{m \leq n} \sum_{d|(m,n)} \mu(d) = \sum_{d|n} \mu(d) \sum_{m \leq n, d|m} 1 = \sum_{d|n} \mu(d) (n/d) \tag{3}$$

Prime Number Theorem (Hadamard, de la Vallee Poussin, 1896): The number $\pi(x)$ of primes below $x \sim x/\log x$ as $x \rightarrow \infty$.

We will show the PNT is "equivalent" to the fact that $\lim_{x \rightarrow \infty} (1/x) \sum_{n \leq x} \mu(n) = 0$.

The later statement has a probabilistic interpretation: If a squarefree integer is chosen at random, then it is equally likely to have an even and an odd number of prime factors.

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The Euler phi (totient) function

Recall that Euler totient function ϕ is defined by

$$\phi(n) := \#\{m : (m, n) = 1, 1 \leq m \leq n\}. \quad (4)$$

Theorem 4.

The Euler phi function satisfies:

1. $\sum_{d|n} \phi(d) = n, \forall n \in \mathbb{N}.$
2. $\phi(n) = \sum_{d|n} \mu(d)(n/d).$
3. $\phi(n)$ is multiplicative.
4. $\phi(n) = \prod_{p^m || n} (p^m - p^{m-1}) = n \prod_{p|n} (1 - 1/p), \forall n \in \mathbb{N}.$

The Euler phi (totient) function

Proof of $\sum_{d|n} \phi(d) = n$, $\forall n \in \mathbb{N}$: Define $\mathbb{Z}_n = 1, \dots, n$,
and for $d, d|n$, $S_d := \{x \in \mathbb{Z}_n : (x, n) = d\}$. We see that $S_d = \{dx' : 1 \leq x' \leq n/d, (x', n/d) = 1\}$.
So $|S_d| = \phi(n/d)$ by definition.
Notice that $\mathbb{Z}_n = \cup_{d:d|n} S_d$, and $S_i \cap S_j = \emptyset$, if $i \neq j$.
So $n = |\mathbb{Z}_n| = \sum_{d|n} |S_d|$.
It follows that $n = \sum_{d|n} \phi(d)$.

$$\phi(n) = \prod_{p^m || n} (p^m - p^{m-1}) = n \prod_{p|n} (1 - 1/p), \forall n \in \mathbb{N}$$

Fact: $\phi(p^m) = p^m - p^{m-1}$.

Hence $\phi(n) = \phi(\prod_{p^m || n} p^m) = \prod_{p^m || n} \phi(p^m) = \prod_{p^m || n} (p^m - p^{m-1}), \forall n \in \mathbb{N}$.

Example 1.

$$\begin{aligned} \phi(1) &= 1, \phi(2) = 2 - 1 = 1, \phi(3) = 3 - 1 = 2, \phi(4) = 2^2 - 2 = 2, \\ \phi(5) &= 5 - 1, \phi(6) = (2 - 1)(3 - 1) = 2, \phi(7) = (7 - 1) = 6. \end{aligned}$$

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Recall that

Von Mangoldt function: $\Lambda(n)$, defined by $\Lambda(n) = 0$ if n is not a prime power, and $\Lambda(p^m) = \log p$ for any prime power p^m .

Theorem 5.

We have

$$\sum_{d|n} \Lambda(d) = \log n \quad (n \in \mathbb{N}). \quad (5)$$

Proof. For $n = 1$, $\Lambda(1) = 0 = \log 1$.

For $n \geq 2$, by definition

$$\sum_{d|n} \Lambda(d) = \sum_{p^m|n} \log p = \log n \quad (6)$$

Motivation: The partial sums $\sum_{n \leq x} \Lambda(n)$ represent a weighted count of the prime powers $p^m \leq x$, with the weights being $\log p$, the correct weights to offset the density of primes.

PNT is equivalent to the assertion that $\lim_{x \rightarrow \infty} (1/x) \sum_{n \leq x} \Lambda(n) = 1$.

Advantage: $\Lambda(n)$ connects to the Riemann zeta function through its Dirichlet series:

$-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)/n^s$. This link enables powerful complex analysis methods, making proofs of PNT more tractable than direct approaches with $\pi(x)$.

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Theorem 1.7. *The divisor function $d(n)$ and the sum-of-divisors function $\sigma(n)$ are multiplicative. Their values at prime powers are given by*

$$d(p^m) = m + 1, \quad \sigma(p^m) = \frac{p^{m+1} - 1}{p - 1}.$$

Definition 4 (Perfect Numbers).

A positive integers n that are equal to the sum of all their proper divisors.

iff condition: $\sigma(n) - n = n$, or $\sigma(n) = 2n$.

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Definition 5.

The Dirichlet product of two arithmetic functions f and g is the arithmetic function defined by

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d) = \sum_{ab=n} f(a)f(b). \quad (7)$$

Motivation 1: form of identities

1. $d(n) = \sum_{d|n} 1 = 1 * 1$
2. $\sigma(n) = \sum_{d|n} d = \text{id} * 1$.
3. $\sum_{d|n} \mu(d) = e(n)$, so $\mu * 1 = e$.
4. $\phi(n) = \sum_{d|n} \mu(d)(n/d)$, so $\phi(n) = \mu * \text{id}$.

Properties of Dirichlet Product

Motivation 2: properties

Theorem 1.8 (Properties of the Dirichlet product).

- (i) *The function e acts as a unit element for $*$, i.e., $f * e = e * f = f$ for all arithmetic functions f .*
- (ii) *The Dirichlet product is commutative, i.e., $f * g = g * f$ for all f and g .*
- (iii) *The Dirichlet product is associative, i.e., $(f * g) * h = f * (g * h)$ for all f, g, h .*
- (iv) *If $f(1) \neq 0$, then f has a unique Dirichlet inverse, i.e., there is a unique function g such that $f * g = e$.*

1. Since $\mu * 1 = e$, the Moebius function is the Dirichlet inverse of the function 1.
2. Since $\phi(n) = \mu * \text{id}$, $\mu * 1 = \mu * \text{id} * 1 = e * \text{id} = \text{id}$, stated in theorem 4.

The Moebius inversion formula

Theorem 6 (Moebius inversion formula).

If $g(n) = \sum_{d|n} f(d)$ for all $n \in \mathbb{N}$, then $f(n) = \sum_{d|n} g(d)\mu(n/d)$ for all n .

Motivation 3: preserve multiplicativity.

Theorem 1.10 (Dirichlet product and multiplicative functions).

- (i) If f and g are multiplicative, then so is $f * g$.*
- (ii) If f is multiplicative, then so is the Dirichlet inverse f^{-1} .*
- (iii) If $f * g = h$ and if f and h are multiplicative, then so is g .*
- (iv) (Distributivity with pointwise multiplication) If h is completely multiplicative, then $h(f * g) = (hf) * (hg)$ for any functions f and g .*

Application I: Proving identities for multiplicative arithmetic functions.

Want to show $(f * g)(n) = h(n)$. Strategy: show $(f * g)(p^m) = h(p^m)$ by characterization of multiplicative function.

1. **Another proof of** $\sum_{d|n} \mu(d) = e(n)$.

Rewrite it as $\mu * 1 = e$, only need to verify that the identity holds on prime powers.

$e(p^m) = 1$, and $(\mu * 1)(p^m) = \sum_{k=0}^m \mu(p^k) = 1 - 1 + 0 - 0 \cdots = 0$, this is indeed the case.

2. **The Dirichlet inverse of λ .** $\lambda(n) = 1$ if $n = 1$, $(-1)^{\sum_{i=1}^k \alpha_i}$ if $n = \prod_{i=1}^k p_i^{\alpha_i}$

To find λ such that $\lambda * f = e$, note f must be multiplicative.

$$0 = e(p^m) = \sum_{k=0}^m f(p^k) \lambda(p^{m-k}) = \sum_{k=0}^m f(p^k) (-1)^{m-k}, \quad (8)$$

So $f(p) = 1$, $f(p^m) = 0$ for $m \geq 2$. $\lambda^{-1} = \mu^2$.

Application II: Evaluating Dirichlet products of multiplicative functions.

the function $\lambda * 1$ $(\lambda * 1)(p^m) = \sum_{k=0}^m \lambda(p^k) = \sum_{k=0}^m (-1)^k = 1$ if m is even, and 0 otherwise.

Application III: Proving the multiplicativity of functions, using known identities.

1. **Multiplicativity of ϕ .** $\phi * 1 = \text{id}$.
2. **Multiplicativity of $d(n)$ and $\sigma(n)$.** $d = 1 * 1$. $\sigma = \text{id} * 1$.